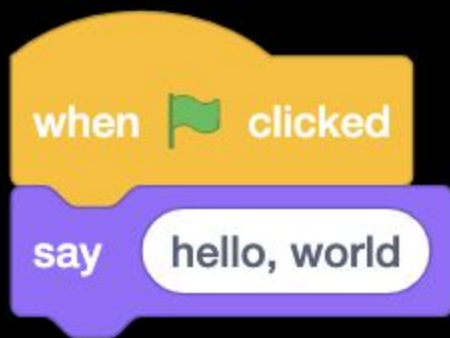


This is CS50



```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    printf("hello, world\n");
```

```
}
```

```
print("hello, world")
```

```
make hello
```

```
./hello
```

```
clang -o hello hello.c -lcs50
```

```
./hello
```

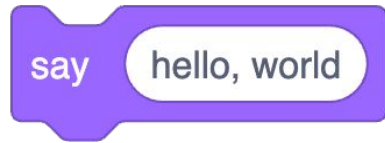
```
python hello.py
```

compiled, interpreted

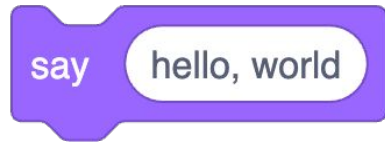
functions







```
printf("hello, world\n");
```



```
print("hello, world")
```

libraries

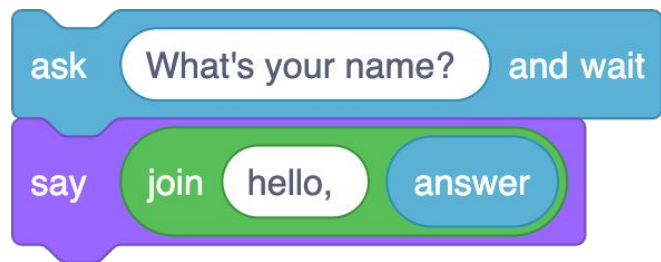
modules, packages

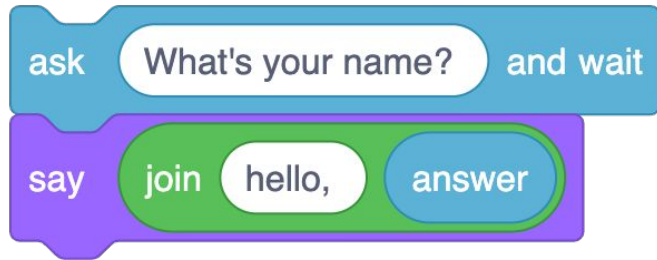
```
#include <cs50.h>
```



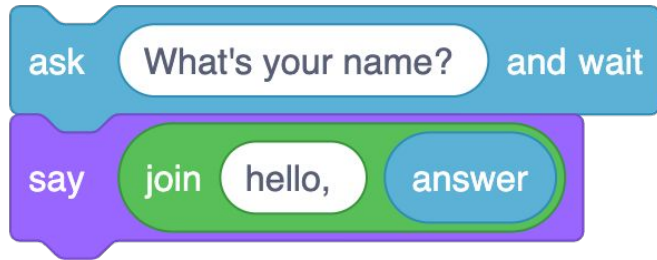
```
import cs50
```

```
from cs50 import get_string
```

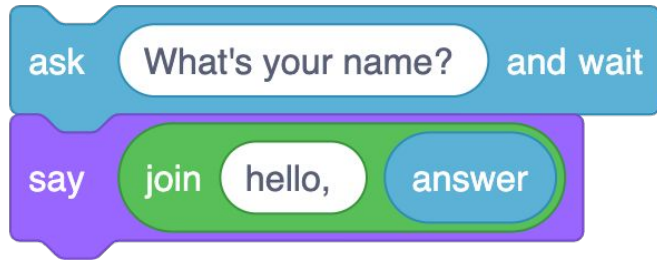




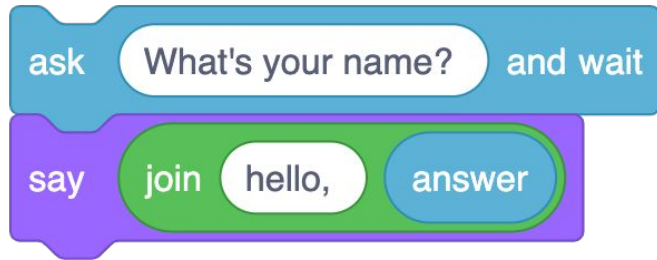
```
string answer = get_string("What's your name? ");  
printf("hello, %s\n", answer);
```



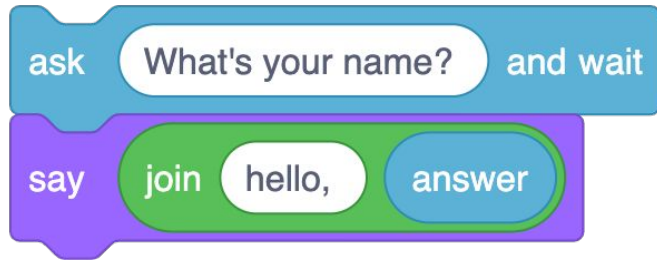
```
answer = get_string("What's your name? ")  
print("hello, " + answer)
```



```
answer = get_string("What's your name? ")  
print("hello,", answer)
```



```
answer = get_string("What's your name? ")  
print(f"hello, {answer}")
```



```
answer = input("What's your name? ")  
print(f"hello, {answer}")
```


positional parameters

named parameters

docs.python.org

docs.python.org/3/library/functions.html

<https://docs.python.org/3/library/functions.html#print>

```
print(*objects, sep=' ', end='\n', file=None, flush=False)
```

```
print(*objects, sep=' ', end='\n', file=None, flush=False)
```

```
print(*objects, sep=' ', end='\n', file=None, flush=False)
```



```
print(*objects, sep=' ', end='\n', file=None, flush=False)
```

variables

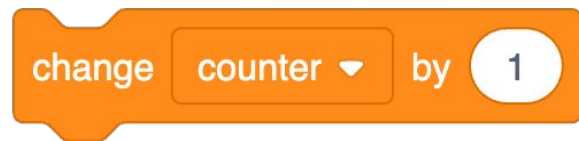


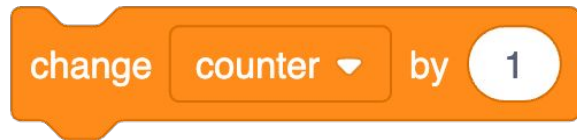


```
int counter = 0;
```

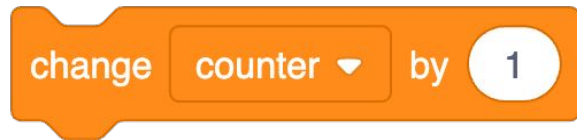


```
counter = 0
```

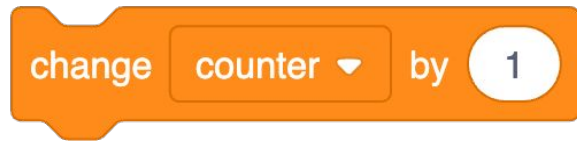




```
counter = counter + 1;
```



```
counter = counter + 1
```

```
counter += 1
```

types

bool

char

double

float

int

long

string

...

bool

float

int

str

...

range

list

tuple

dict

set

...

get_char

get_double

get_float

get_int

get_long

get_string

...

get_float

get_int

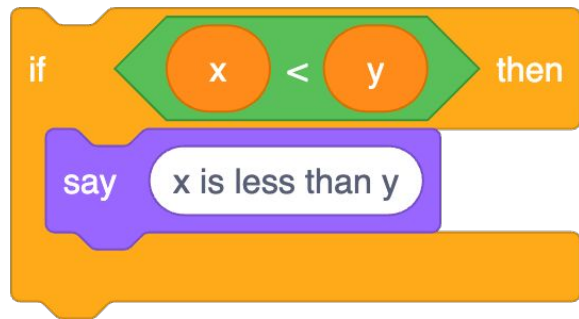
get_string

```
from cs50 import get_float  
from cs50 import get_int  
from cs50 import get_string
```



```
from cs50 import get_float, get_int, get_string
```

conditionals

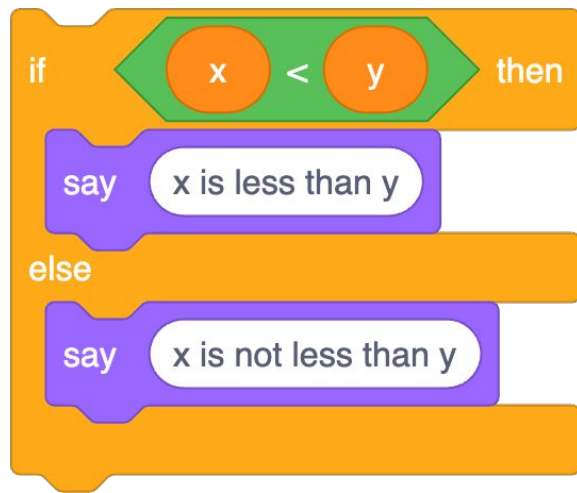


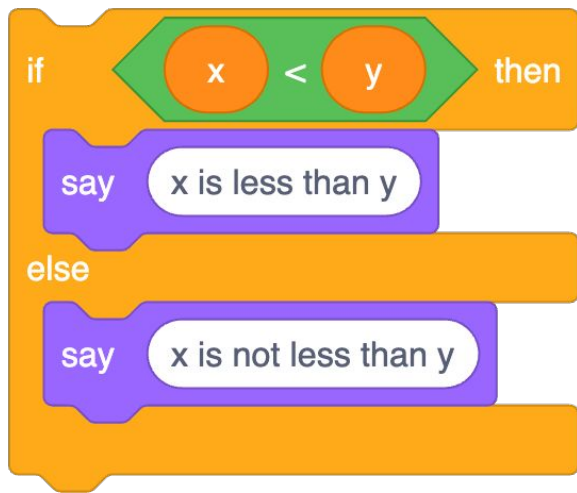


```
if (x < y)
{
    printf("x is less than y\n");
}
```

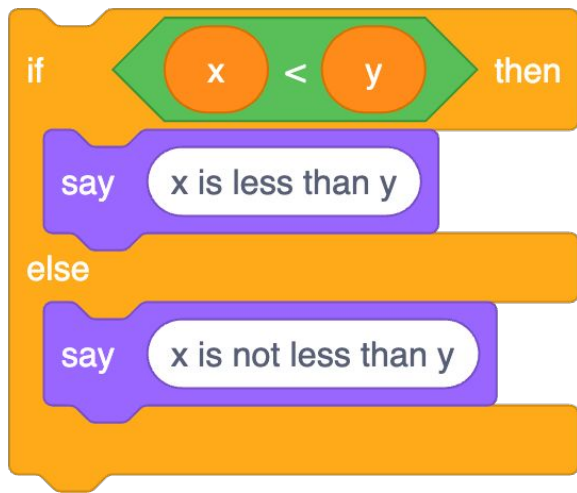


```
if x < y:  
    print("x is less than y")
```

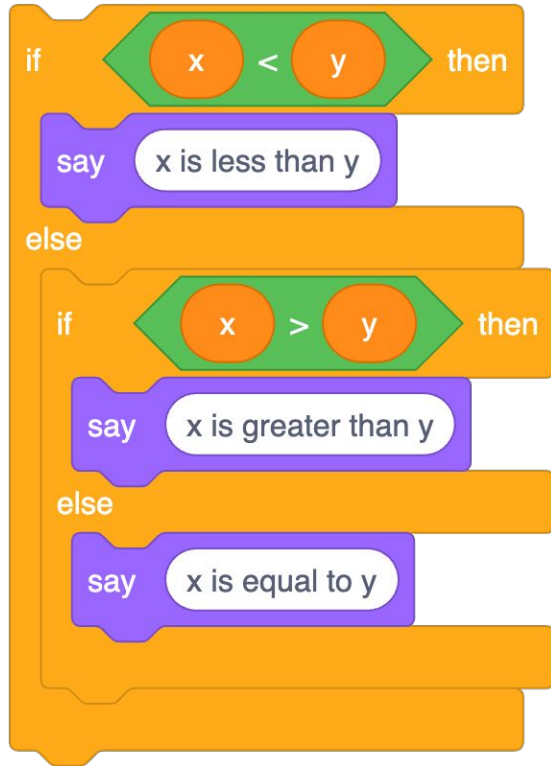


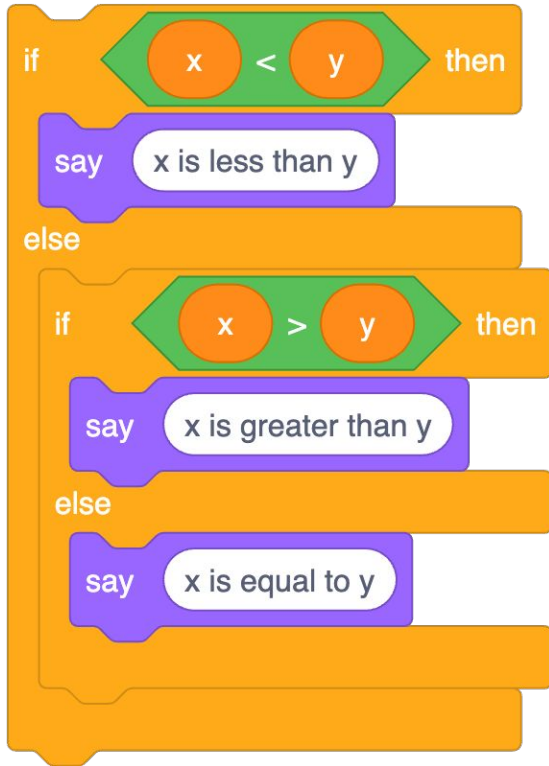


```
if (x < y)
{
    printf("x is less than y\n");
}
else
{
    printf("x is not less than y\n");
}
```

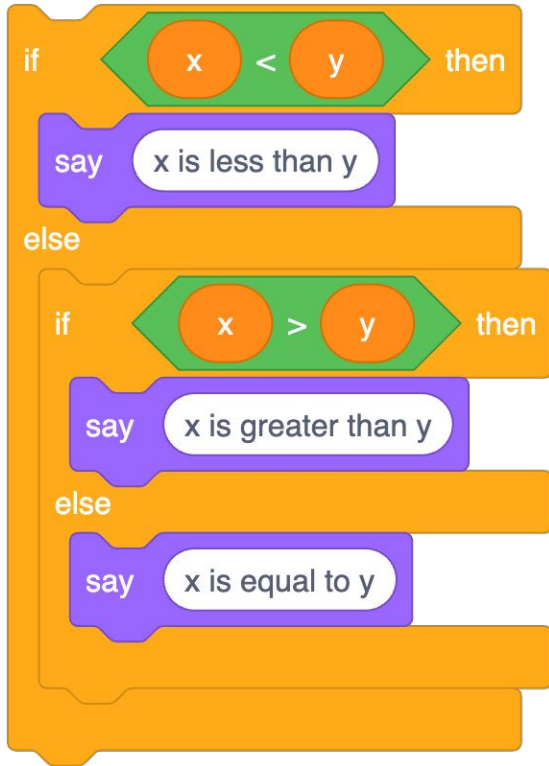


```
if x < y:  
    print("x is less than y")  
else:  
    print("x is not less than y")
```



```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else
{
    printf("x is equal to y\n");
}
```



```
if x < y:  
    print("x is less than y")  
elif x > y:  
    print("x is greater than y")  
else:  
    print("x is equal to y")
```

str

list

docs.python.org/3/library/stdtypes.html#sequence-types-list-tuple-range

object-oriented programming

OOP

docs.python.org/3/library/stdtypes.html#string-methods

loops





```
int i = 0;  
while (i < 3)  
{  
    printf("meow\n");  
    i++;  
}
```



```
i = 0
while i < 3:
    print("meow")
    i += 1
```





```
for (int i = 0; i < 3; i++)  
{  
    printf("meow\n");  
}
```



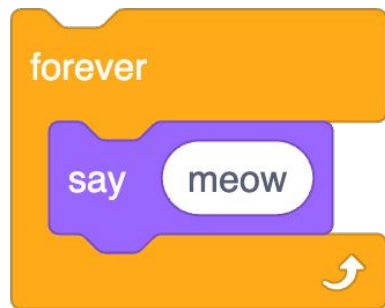
```
for i in [0, 1, 2]:  
    print("meow")
```

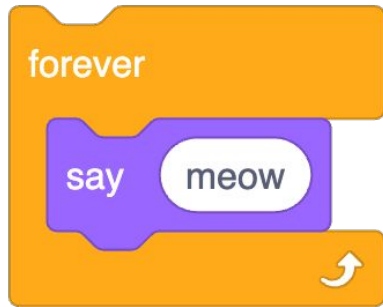



```
for i in range(3):  
    print("meow")
```

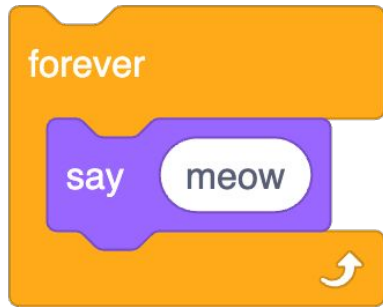


```
for _ in range(3):  
    print("meow")
```





```
while (true)
{
    printf("meow\n");
}
```



```
while True:  
    print("meow")
```

GOOD SOURCE OF
VITAMIN C
GLUTEN FREE

48
ROLLS

FRUIT BY THE
Foot

TM FRUIT
FLAVORED
SNACKS

BERRY TIE-DYE™
FLAVORED WITH OTHER NATURAL FLAVORS

STRAWBERRY TIE-DYE™
FLAVORED WITH OTHER NATURAL FLAVORS

ENLARGED TO
SHOW DETAIL

48 - 0.75 OZ (21g) ROLLS
NET WT 2 LB 4 OZ (36 OZ) (1.02kg)

PER ROLL

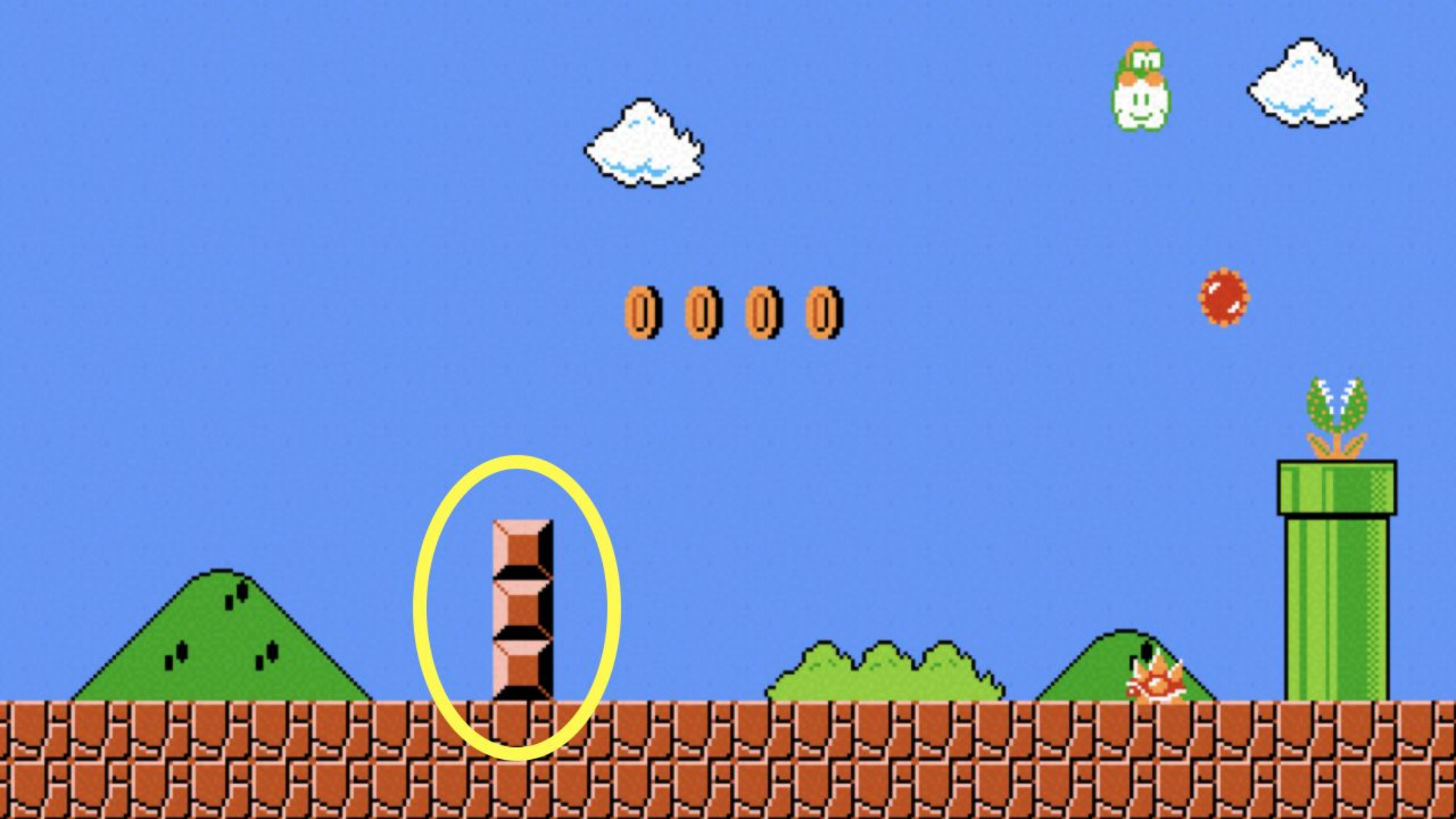
80 CALORIES	0.5g SAT FAT 2% DV	50mg SODIUM 2% DV	9g ADDED SUGARS 17% DV
-----------------------	---------------------------------	--------------------------------	--

truncation

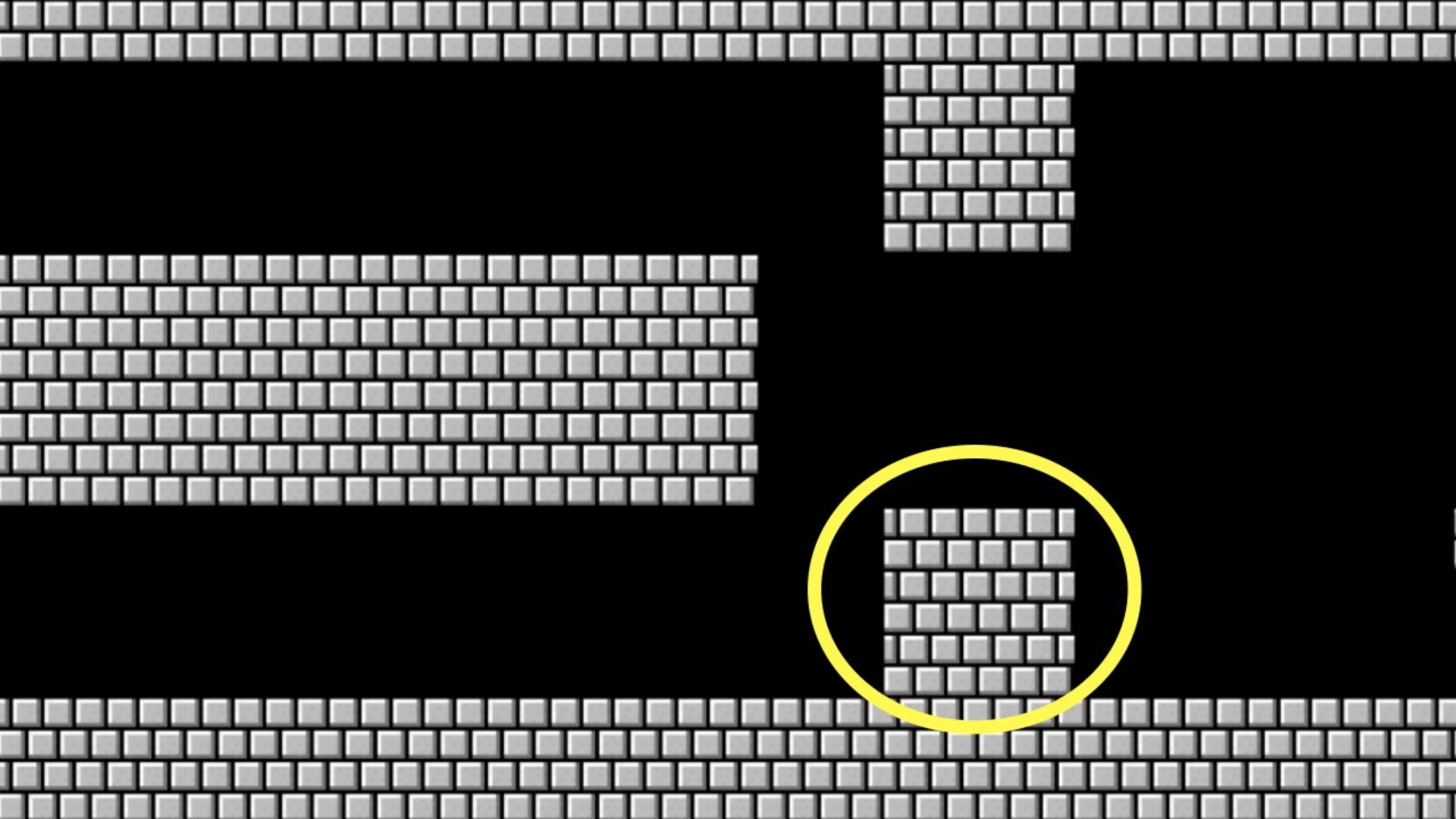
floating-point imprecision

integer overflow

exceptions







len

docs.python.org/3/library/functions.html#len

dict

key	value

docs.python.org/3/library/stdtypes.html#mapping-types-dict

phonebook - Google Sheets

https://docs.google.com/spreadsheets/

Guest

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CS50 | *fx*

	A	B	C	D	E	F	G	
1	name	number						
2	Kelly	+1-617-495-1000						
3	David	+1-617-495-1000						
4	John	+1-949-468-2750						
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								

+

≡

Sheet1

sys

docs.python.org/3/library/sys.html

CSV

docs.python.org/3/library/csv.html

pip

This is CS50